

8.421 Atomic Physics I, Spring 2025

Problem Set 5

Prof. Martin Zwierlein

March 25, 2025

1 Symmetries and Permanent Dipole Moments

The molecule HCl is reported to have a permanent electric dipole moment (EDM) of $0.41\ e a_o$. There is more than one way to answer each of the three parts.

a) Considering electromagnetic interactions only, the Hamiltonian of HCl is rotationally invariant in the lab frame. Explain the connection between this and HCl's observed electric dipole moment. (Hint: This question can be answered by thinking about both parity and rotational invariance.)

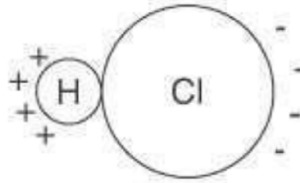


Figure 1: HCl molecule

A chlorine atom is said to be more electronegative than a hydrogen atom; in the HCl molecule, one finds that the distribution of electrons is pulled toward the chlorine atom, as indicated in the picture above. When we have a fixed molecule like this, we see that it makes sense to say that HCl has a permanent EDM. When chemists speak of the dipole moment of a molecule, they are referring to the dipole moment in this rest frame.

In the lab frame, however, the energy of the electromagnetic interactions of the nuclei and electrons is invariant under rotations of the whole molecule. This means that the total angular momentum J commutes with the Hamiltonian; J and m_J are thus good quantum numbers, and the corresponding eigenstates must have definite parity. Since $\mathbf{r}$ is an odd operator, we know $\langle J m_J | \mathbf{r} | J m_J \rangle = 0$, and it follows that the expectation value of the electric dipole operator is always zero. This means there is no EDM in the lab frame. Another way to think about this: in the absence of an applied electric field, the internuclear axis of the molecule is equally likely to point in any direction.

As noted above, the EDM is clearly nonzero in the frame where the molecule is at rest. In this frame, the Hamiltonian is not invariant under general rotations; one must add extra non-invariant terms to the molecule's Hamiltonian in order to fix the orientation of its axis. The eigenstates do not have definite parity, and it is indeed possible to have a nonzero expectation value for the dipole operator. Another way to see this: to construct a state where the molecule is pointing in a definite direction, one must make a superposition of angular momentum eigenstates; such a superposition will not have definite parity.

b) The fact that a “permanent” EDM value is quoted for HCl indicates that there is an experimental regime where the observed dipole moment is independent of the electric field. This can be observed as a

linear Stark effect. Given the parity considerations from part (a), where does this measured linear Stark effect come from?

To find the shifts of the energy levels in HCl as a function of an applied electric field $\mathbf{E}_o$, we treat the Stark Hamiltonian $H' = e\mathbf{E}_o \cdot \mathbf{r} = eE_oz$ as a perturbation. Suppose we want to find the shift due to H' of the eigenstate $|\psi_i\rangle$ (with corresponding energy E_i) in the lab frame. We already know that $\langle\psi_i|H'|\psi_i\rangle = 0$, so we need to apply perturbation theory of at least second order, which involves the matrix elements of H' with other eigenstates of opposite parity $\{|\psi_j\rangle\}$. As long as $|\langle\psi_i|H'|\psi_j\rangle| \ll |E_i - E_j|$ for all j , second-order perturbation theory predicts a quadratic Stark effect. The energy shift in this case turns out to be

$$\frac{I_o d^2}{\hbar^2} \left[\frac{J(J+1) - 3m_J^2}{J(J+1)(2J-1)(2J+3)} \right] E_o^2, \quad (1)$$

where I_o is the moment of inertia and d is the permanent EDM.

If the field is large enough such that $|\langle\psi_i|H'|\psi_j\rangle| > |E_i - E_j|$ for some j , then second-order perturbation theory is no longer valid. At still larger fields, degenerate perturbation theory applies, which amounts to diagonalizing H' in the basis of “nearly-degenerate” eigenstates. Suppose there is one state $|\psi_j\rangle$ much closer in energy to $|\psi_i\rangle$ than any other. The shifts of the energy levels turn out to be linear. Also, there is maximal mixing of the states $|\psi_i\rangle$ and $|\psi_j\rangle$. The new eigenstates are approximately

$$|\psi_{\pm}\rangle = \frac{1}{\sqrt{2}}(|\psi_i\rangle \pm |\psi_j\rangle). \quad (2)$$

Because these mixed states are a superposition of opposite-parity states, they will generally exhibit an EDM. In fact, since these states are maximally mixed, they represent the states with the largest possible EDM for a given $|\psi_i\rangle$. The linear Stark effect is, then, equivalent to inducing the largest possible induced EDM in a molecule. This regime can generally be reached by applying a sufficiently large electric field.

The fact that a “permanent” EDM value for HCl is quoted indicates that there is an experimental regime where the observed dipole moment is independent of the electric field. In this regime, HCl has a linear Stark effect. As it turns out, it doesn’t require a very strong electric field to reach this regime, since the “ground state” of HCl consists of two nearly-degenerate states of opposite parity. Because of this, some would say HCl effectively has a permanent EDM even in the lab frame: as soon as an external field is turned on, one observes the characteristic dipole moment, $0.41ea_o$.

c) Justify why atoms can have permanent magnetic dipole moments, but not EDM’s. A clearly explained diagram will suffice.

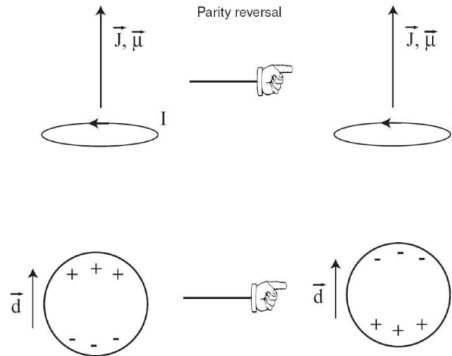


Figure 2: Magnetic and electric dipoles under parity reversal.

For an atom, there is no difference in the Hamiltonian between lab frame and rest frame. An atomic eigenstate has definite parity, and it cannot exhibit an EDM. Another way to look at this is to consider

what happens when an atom is subjected to a parity reversal operation: the atom's angular momentum is unchanged, but any EDM would be reversed in direction (see diagram below). Thus, the existence of an EDM constitutes a parity violation. On the other hand, the magnetic moment of an atom is always proportional to its angular momentum. A magnetic moment does not reverse direction under a parity operation, and it is consistent with parity symmetry; it is reasonable to expect atoms to have permanent magnetic moments.

Vector quantities which transform to their inverse under a parity operation are known as polar vectors. And vectors which remain unchanged under a parity operation are called axial vectors. Thus, an EDM is a polar vector, and a magnetic dipole moment is an axial vector. For a more precise discussion of polar vectors, axial vectors, and the parity operation (spatial inversion), take a look at Jackson's Classical Electrodynamics, 2nd Ed., pp. 247–249.

Finally, it is interesting to note that some extensions to the Standard Model allow for a permanent EDM of the electron which could lead to tiny ($< 10^{-16}ea_0$) permanent atomic EDM's. A number of impressive experiments have been undertaken in the past two decades to look for an electron EDM in atoms and molecules. The latest of these experiments are stringent enough to test the predictions of Standard Model extensions. So far, all experimental results are consistent with zero permanent EDM in atomic systems.

2 Spherical Harmony

Our goal is to be able to evaluate matrix elements like

$$\langle J'm'_J | Y_{Lm} | Jm_J \rangle = \int d\Omega Y_{J'm'_J}^* Y_{Lm} Y_{Jm_J} \quad (3)$$

of which $L = 1$ is relevant for dipole transitions, $L = 2$ for quadrupole etc...

To this end, consider two particles with angular momenta j_1 and j_2 . The total angular momentum is $\mathbf{J} = \mathbf{j}_1 + \mathbf{j}_2$. We know that we can couple the states $|j_1 m_1\rangle$ and $|j_2 m_2\rangle$ into states of definite total angular momentum J with the help of Clebsch-Gordan coefficients:

$$|(j_1 j_2) JM\rangle = \sum_{m_1, m_2} |j_1 m_1\rangle |j_2 m_2\rangle \langle j_1 m_1 j_2 m_2 | JM \rangle \quad (4)$$

$$|j_1 m_1\rangle |j_2 m_2\rangle = \sum_{J, M} |(j_1 j_2) JM\rangle \langle JM | j_1 m_1 j_2 m_2 \rangle \quad (5)$$

The Clebsch-Gordan coefficients imply that the sum over M has just one non-zero term, $M = m_1 + m_2$, and that J runs from $|j_1 - j_2|$ to $j_1 + j_2$. The wavefunction of each particle at polar angle $(i, \phi_i) \equiv \Omega_i$ is $\langle \Omega_i | j_i m_i \rangle = Y_{j_i m_i}(\Omega_i)$ for $i = 1, 2$, and that for the state of definite total angular momentum is $\Phi_{JM}(\Omega_1, \Omega_2) = \langle \Omega_1, \Omega_2 | (j_1 j_2) JM \rangle$. Note that the latter requires two sets of polar angles. The function $F_{JM}(\Omega) \equiv \langle \Omega, \Omega | (j_1 j_2) JM \rangle$, where the two polar angles $\Omega_1 = \Omega_2 = \Omega$ are equal, is a wavefunction of an effective particle with angular momentum quantum numbers J, M . Indeed, it inherits its eigenvalues of J^2 and J_z from $\Phi_{JM}(\Omega_1, \Omega_2)$. So $F_{JM}(\Omega)$ must be proportional to the spherical harmonic $Y_{JM}(\Omega)$:

$$F_{JM}(\Omega) = A_{(j_1 j_2)J} Y_{JM}(\Omega) \quad (6)$$

The factor $A_{(j_1 j_2)J}$ cannot depend on M as F_{JM} must behave in all respects like Y_{JM} , in particular when acted upon with $J_{\pm}$, which changes M . So we have shown a relation for the spherical harmonics:

$$A_{(j_1 j_2)J} Y_{JM}(\Omega) = \sum_{m_1, m_2} \langle j_1 m_1 j_2 m_2 | JM \rangle Y_{j_1 m_1}(\Omega) Y_{j_2 m_2}(\Omega) \quad (7)$$

YOUR TASKS:

a) Find $A_{(j_1 j_2)J}$.

Hint: Recall that $Y_{lm}(\theta = 0, \phi) = \sqrt{\frac{2l+1}{4\pi}} \delta_{m0}$.

The relation above should be equal for all solid angles, so it is clever (also looking at the hint) to consider a particular direction, namely with the polar angle $\Omega_0 = (\theta, \phi) = (0, \phi)$ pointing in the z -direction.

So plugging that solid angle into 2 gives

$$F_{JM}(\Omega_0) = \langle \Omega_0, \Omega_0 | (j_1 j_2) JM \rangle \quad (8)$$

$$= \sum_{m_1, m_2} \langle \Omega_0 | j_1, m_1 \rangle \langle \Omega_0 | j_2, m_2 \rangle \langle j_1 m_1 j_2 m_2 | JM \rangle \quad (9)$$

$$= \sum_{m_1, m_2} Y_{j_1 m_1}(\theta = 0, \phi) Y_{j_2, m_2}(\theta = 0, \phi) \langle j_1 m_1 j_2 m_2 | JM \rangle \quad (10)$$

$$= \sum_{m_1, m_2} \sqrt{\frac{2j_1 + 1}{4\pi}} \delta_{m_1, 0} \sqrt{\frac{2j_2 + 1}{4\pi}} \delta_{m_2, 0} \langle j_1 m_1 j_2 m_2 | JM \rangle \quad (11)$$

$$= \frac{\sqrt{(2j_1 + 1)(2j_2 + 1)}}{4\pi} \langle j_1 0 j_2 0 | JM \rangle \quad (12)$$

$$= \frac{\sqrt{(2j_1 + 1)(2j_2 + 1)}}{4\pi} \langle j_1 0 j_2 0 | J0 \rangle \delta_{M, 0} \quad (13)$$

where the last $\delta_{M, 0}$ is a consequence of the Clebsch-Gordan coefficient requiring $M = m_1 + m_2 = 0$ (this is just z -angular momentum conservation). We also have

$$F_{JM}(\Omega_0) = A_{(j_1 j_2)J} Y_{JM}(\Omega_0) \quad (14)$$

$$= A_{(j_1 j_2)J} \sqrt{\frac{2J + 1}{4\pi}} \delta_{M, 0} \quad (15)$$

So therefore:

$$A_{(j_1 j_2)J} = \sqrt{\frac{(2j_1 + 1)(2j_2 + 1)}{4\pi(2J + 1)}} \langle j_1 0 j_2 0 | J0 \rangle \quad (16)$$

So the relation 5 is

$$Y_{JM}(\Omega) = \sqrt{\frac{4\pi(2J + 1)}{(2j_1 + 1)(2j_2 + 1)}} \sum_{m_1, m_2} \frac{\langle j_1 m_1 j_2 m_2 | JM \rangle}{\langle j_1 0 j_2 0 | J0 \rangle} Y_{j_1 m_1}(\Omega) Y_{j_2 m_2}(\Omega) \quad (17)$$

We now have an appreciation for why sometimes one prefers to work with the modified spherical harmonics

$$C_{JM}(\Omega) \equiv \sqrt{\frac{4\pi}{2J + 1}} Y_{JM}(\Omega) \quad (18)$$

which at the north pole are simply $C_{JM}(\theta = 0, \phi) = \delta_{M, 0}$. The expression is then written more simply:

$$C_{JM}(\Omega) = \sum_{m_1, m_2} \frac{\langle j_1 m_1 j_2 m_2 | JM \rangle}{\langle j_1 0 j_2 0 | J0 \rangle} C_{j_1 m_1}(\Omega) C_{j_2 m_2}(\Omega) \quad (19)$$

b) Find an expression relating $Y_{j_1 m_1}(\Omega) Y_{j_2 m_2}(\Omega)$ to a sum over the $Y_{JM}(\Omega)$. Applying $\langle \Omega, \Omega |$ to the left of Eq. 3 gives, according to the definition of $F_{JM}(\Omega)$:

$$Y_{j_1 m_1}(\Omega) Y_{j_2 m_2}(\Omega) = \sum_{JM} F_{JM}(\Omega) \langle JM | j_1 m_1 j_2 m_2 \rangle \quad (20)$$

Using $F_{JM}(\Omega) = A_{(j_1 j_2)J} Y_{JM}(\Omega)$ yields

$$Y_{j_1 m_1}(\Omega) Y_{j_2 m_2}(\Omega) = \sum_{JM} A_{(j_1 j_2)J} \langle JM | j_1 m_1 j_2 m_2 \rangle Y_{JM}(\Omega) \quad (21)$$

$$= \sum_{JM} \sqrt{\frac{(2j_1+1)(2j_2+1)}{4\pi(2J+1)}} \langle j_1 0 j_2 0 | J 0 \rangle \langle JM | j_1 m_1 j_2 m_2 \rangle Y_{JM}(\Omega) \quad (22)$$

Again, using modified spherical harmonics turns this into the simpler

$$C_{j_1 m_1}(\Omega) C_{j_2 m_2}(\Omega) = \sum_{JM} \langle j_1 0 j_2 0 | J 0 \rangle \langle JM | j_1 m_1 j_2 m_2 \rangle C_{JM}(\Omega) \quad (23)$$

It makes of course good sense that it is possible to express the product of two spherical harmonics in terms of a linear superposition of single Y_{JM} 's. Any function of the solid angle Ω can be decomposed into the spherical harmonics, since they provide a complete basis of functions on the unit sphere.

c) Find the matrix element $\langle j_3 m_3 | Y_{j_2 m_2} | j_1 m_1 \rangle$.

We would like to calculate

$$\langle j_3 m_3 | Y_{j_2 m_2} | j_1 m_1 \rangle = \int d\Omega Y_{j_3 m_3}^*(\Omega) Y_{j_2 m_2}(\Omega) Y_{j_1 m_1}(\Omega) \quad (24)$$

We can use our result for the product of two spherical harmonics in terms of sums of individual ones for the last two factors (for example). Then we see

$$\langle j_3 m_3 | Y_{j_2 m_2} | j_1 m_1 \rangle = \sum_{JM} A_{(j_1, j_2)J} \int d\Omega Y_{j_3 m_3}^*(\Omega) Y_{JM}(\Omega) \langle JM | j_2 m_2 j_1 m_1 \rangle \quad (25)$$

$$= A_{(j_1, j_2)j_3} \langle j_3 m_3 | j_2 m_2 j_1 m_1 \rangle \quad (26)$$

Combining this with our result for $A_{(j_1, j_2)j_3}$ gives

$$\langle j_3 m_3 | Y_{j_2 m_2} | j_1 m_1 \rangle = \sqrt{\frac{(2j_1+1)(2j_2+1)}{4\pi(2j_3+1)}} \langle j_1 0 j_2 0 | j_3 0 \rangle \langle j_3 m_3 | j_2 m_2 j_1 m_1 \rangle \quad (27)$$

This is an example of the Wigner-Eckart theorem: The $Y_{JM}(\Omega)$ are tensors of rank J , and matrix elements of tensors factorize into a directional part - the Clebsch-Gordan coefficient - and a scalar. Physically, we see that for the matrix element Eq. 22 to be non-zero, the integrand must be a scalar. So the only part of the state $Y_{j_2 m_2} | j_1 m_1 \rangle$ that can contribute is that which rotates like $| j_3 m_3 \rangle$. By definition, this amplitude is given by the Clebsch-Gordan coefficient, and this is the geometrical factor containing all dependence on magnetic quantum numbers.

We can directly read off selection rules:

- i) We have $m_3 = m_1 + m_2$, which we can interpret this as conservation of z angular momentum. It is directly obvious by considering the ϕ -integral, which is $\int_0^{2\pi} d\phi e^{i(m_1+m_2-m_3)\phi} = \delta_{0, m_1+m_2-m_3}$.
- ii) We must be able to form a triangle of lengths j_1 , j_2 and j_3 (so j_3 must lie between $|j_2 - j_1|$ and $j_1 + j_2$).
- iii) In addition, the prefactor $\langle j_1 0 j_2 0 | j_3 0 \rangle$ implies an additional rule: $j_1 + j_2 - j_3$ must be even. This is just expressing the parity rule: For the integral over the unit sphere to be non-zero, the product of the three spherical harmonics must be an even function under parity.

3 Dipole operator

A symmetric top molecule has a Hamiltonian $H = BJ^2$, with B the rotational constant. The dipole moment operator is $\hat{\mathbf{d}} = d\hat{\mathbf{r}}$, with d the value of the "permanent dipole moment" (in the molecular frame)

a) Show that the unit vector $\hat{\mathbf{r}}$ can be written in terms of the modified spherical harmonics $C_{1m}(\theta, \phi) = \sqrt{\frac{4\pi}{3}}Y_{1m}(\theta, \phi)$, the unit vector $\hat{\mathbf{e}}_z \equiv \hat{\mathbf{e}}_0$ and the vectors

$$\hat{\mathbf{e}}_+ \equiv \hat{\mathbf{e}}_{+1} \equiv -\frac{\hat{\mathbf{e}}_x + i\hat{\mathbf{e}}_y}{\sqrt{2}} \quad (28)$$

$$\hat{\mathbf{e}}_- \equiv \hat{\mathbf{e}}_{-1} \equiv \frac{\hat{\mathbf{e}}_x - i\hat{\mathbf{e}}_y}{\sqrt{2}} \quad (29)$$

as what's called the spherical tensor decomposition:

$$\hat{\mathbf{r}} = \sum_m C_{1m}^* \hat{\mathbf{e}}_m = \sum_m C_{1m} \hat{\mathbf{e}}_m^* \quad (30)$$

We have

$$\hat{\mathbf{r}} = \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} \quad (31)$$

The spherical harmonics of order $l = 1$ are

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta \quad (32)$$

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} \quad (33)$$

$$Y_{1-1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi} \quad (34)$$

The modified spherical harmonics $C_{1m}(\theta, \phi) = \sqrt{\frac{4\pi}{3}}Y_{1m}(\theta, \phi)$ are then

$$C_{10} = \cos \theta = \hat{z} \quad (35)$$

$$C_{11} = -\frac{1}{\sqrt{2}} \sin \theta e^{i\phi} = -\frac{\hat{x} + i\hat{y}}{\sqrt{2}} = -C_{1-1}^* \quad (36)$$

$$C_{1-1} = \frac{1}{\sqrt{2}} \sin \theta e^{-i\phi} = \frac{\hat{x} - i\hat{y}}{\sqrt{2}} = -C_{11}^* \quad (37)$$

One can write

$$\hat{\mathbf{r}} = \hat{x}\hat{\mathbf{e}}_x + \hat{y}\hat{\mathbf{e}}_y + \hat{z}\hat{\mathbf{e}}_z \quad (38)$$

$$= \frac{\hat{x} + i\hat{y}}{\sqrt{2}} \frac{\hat{\mathbf{e}}_x - i\hat{\mathbf{e}}_y}{\sqrt{2}} + \frac{\hat{x} - i\hat{y}}{\sqrt{2}} \frac{\hat{\mathbf{e}}_x + i\hat{\mathbf{e}}_y}{\sqrt{2}} + \hat{z}\hat{\mathbf{e}}_z \quad (39)$$

$$= -C_{11}\hat{\mathbf{e}}_- - C_{1-1}\hat{\mathbf{e}}_+ + C_{10}\hat{\mathbf{e}}_0 \quad (40)$$

$$= C_{11}^*\hat{\mathbf{e}}_+ + C_{1-1}^*\hat{\mathbf{e}}_- + C_{10}\hat{\mathbf{e}}_0 \quad (41)$$

$$= \sum_m C_{1m}^* \hat{\mathbf{e}}_m = \sum_m C_{1m} \hat{\mathbf{e}}_m^* \quad (42)$$

b) Show that

$$\hat{\mathbf{e}}_m^* \cdot \hat{\mathbf{e}}_n = \sum_p \delta_{mp} \delta_{np} = \delta_{mn} \quad (43)$$

(so in particular $\hat{\mathbf{e}}_+^* \cdot \hat{\mathbf{e}}_- = 0$, $\hat{\mathbf{e}}_-^* \cdot \hat{\mathbf{e}}_+ = 0$ and $\hat{\mathbf{e}}_+^* \cdot \hat{\mathbf{e}}_+ = \hat{\mathbf{e}}_-^* \cdot \hat{\mathbf{e}}_- = 1$).

A few steps of algebra gives the result.

c) As a by-product of this formalism, by taking two unit vectors $\hat{\mathbf{r}}$ and $\hat{\mathbf{r}}'$, pointing in the direction of solid angle (θ, ϕ) and (θ', ϕ') , derive a known fact from geometry, namely

$$\cos(\Theta) = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi') \quad (44)$$

where Θ is the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{r}}'$, so between (θ, ϕ) and (θ', ϕ') .

This is a very fancy way to write simply $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}' = \cos(\Theta)$. The result is of course immediate if we use the components of $\hat{\mathbf{r}}$ and $\hat{\mathbf{r}}'$, as we get $\cos \theta \cos \theta' + \sin \theta \sin \theta' (\cos \phi \cos \phi' + \sin \phi \sin \phi')$ and, using a trig identity, the desired result. Instead, here we are tasked to make use of the spherical tensor decomposition $\hat{\mathbf{r}} = \sum_m C_{1m}^* \hat{\mathbf{e}}_m$ and $\hat{\mathbf{r}}' = \sum_m C_{1m}'^* \hat{\mathbf{e}}_m$.

We have

$$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}' = \sum_{m,m'} C_{1m} C_{1m'}'^* \hat{\mathbf{e}}_m^* \cdot \hat{\mathbf{e}}_{m'} \quad (45)$$

$$= \sum_{m,m'} C_{1m} C_{1m'}'^* \delta_{m,m'} \quad (46)$$

$$= \sum_m C_{1m} C_{1m}'^* \quad (47)$$

$$= \cos \theta \cos \theta' + \frac{\sin \theta}{\sqrt{2}} e^{i\phi} \frac{\sin \theta'}{\sqrt{2}} e^{-i\phi'} + \frac{\sin \theta}{\sqrt{2}} e^{-i\phi} \frac{\sin \theta'}{\sqrt{2}} e^{i\phi'} \quad (48)$$

$$= \cos \theta \cos \theta' + \sin \theta \sin \theta' \frac{1}{2} \left(e^{i(\phi-\phi')} + e^{-i(\phi-\phi')} \right) \quad (49)$$

$$= \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi') \quad (50)$$

Additional information for general knowledge: In fact, the equation is the $l = 1$ version of the general

$$P_l(\cos \Theta) = \sum_m C_{lm}^*(\theta, \phi) C_{lm}(\theta', \phi') \quad (51)$$

with $C_{lm}(\theta, \phi) = \sqrt{\frac{4\pi}{2l+1}} Y_{lm}$. This fact we used in multipole decomposition of the Coulomb interaction between electrons and nucleus.

The proof for general l is as follows. The scalar product $\mathbf{C}_l \cdot \mathbf{C}_l' \equiv \sum_m C_{lm}^*(\theta, \phi) C_{lm}(\theta', \phi')$ is invariant under rotations of axes. It can thus only be a function of the angle Θ between the directions (θ, ϕ) and (θ', ϕ') , this angle being the only quantity independent of the choice of axes. Choosing axes so that (θ', ϕ') becomes the new z -axis we have $C_{lm}(\theta', \phi') \rightarrow C_{lm}(0, 0) = \delta_{m0}$. And $C_{l0}(\theta, \phi) \rightarrow P_l(\cos \theta) = P_l(\cos \Theta)$.

d) The electric field can be written

$$\mathbf{E} = E_z \hat{\mathbf{e}}_z + E_x \hat{\mathbf{e}}_x + E_y \hat{\mathbf{e}}_y \quad (52)$$

$$= E_0 \hat{\mathbf{e}}_0 + E_-^* \hat{\mathbf{e}}_- + E_+^* \hat{\mathbf{e}}_+ \quad (53)$$

$$= \sum_m E_m^* \hat{\mathbf{e}}_m = \sum_m E_m \hat{\mathbf{e}}_m^* \quad (54)$$

with $E_0 = E_z$, $E_+ = -\frac{1}{\sqrt{2}}(E_x + iE_y)$, $E_- = \frac{1}{\sqrt{2}}(E_x - iE_y)$. Show that the dipole operator can be written

$$-\hat{\mathbf{d}} \cdot \mathbf{E} = -d \sum_m C_{1m}^* E_m = -d \sum_m C_{1m} E_m^* \quad (55)$$

This is a similar exercise in algebra from above. Writing it this way makes it clear which combination of electric fields are coupled to which operator, and so which combination causes what change in magnetic quantum number.

e) Extra credit: Take $\mathbf{E} = E\hat{z}$. So the only interaction is $\propto C_{10}$. In Mathematica (or your preferred program) setup the Hamiltonian matrix $H = BJ^2 - \hat{\mathbf{d}} \cdot \mathbf{E}$, including states up to sufficiently high J to give the energies of the first six energy levels (up to $|2, 0\rangle$) up to an electric field $E \approx 10B/d$. Plot the probability $|\langle \theta, \phi | \Psi \rangle|^2$ (using e.g. SphericalPlot3D) of the lowest state ($|0, 0\rangle$ at $E = 0$) for a few values, e.g. $dE/B = 0, 1, 10$.

Additional information for general knowledge: For matrix elements of the dipole operator we have

$$\langle J'm'_J | C_{1m} | Jm_J \rangle = \int d\Omega Y_{J'm'_J}^* C_{1m} Y_{Jm_J} \quad (56)$$

which you calculated in problem 1. From this follow the selection rules $\Delta J = \pm 1$ and $\Delta M = 0, \pm 1$. Note that $C_{11} \propto Y_{11}$ increases the magnetic quantum number by $\Delta M = +1$, while $C_{1,-1}$ lowers it by $\Delta M = -1$. $C_{1,0}$ gives $\Delta M = 0$.

The somewhat surprising choices of signs in the components E_+ and E_- are made so that the E_m transform under rotations as irreducible tensors of rank 1. If D is the operator of an arbitrary rotation (e.g. described by Euler angles), then an irreducible tensor of rank k has $2k+1$ components T_{kq} which transform as

$$T'_{kq} = DT_{kq}D^\dagger = \sum_p T_{kp} D_{pq}^k(\alpha\beta\gamma).$$

with $D_{pq}^k(\alpha\beta\gamma)$ the entries of the rotation (Wigner D-)matrix for given Euler angles. Writing this for an infinitesimal rotation $D = 1 - i\alpha J_\lambda$ by angle α about an axis λ yields commutation relations of T_{kq} with the angular momentum operators that each irreducible tensor must obey. In particular, for J_z and $J_\pm$ one finds

$$[J_z, T_{kq}] = qT_{kq} \quad (57)$$

$$[J_\pm, T_{kq}] = \sqrt{k(k+1) - q(q \pm 1)} T_{kq \pm 1} \quad (58)$$

These commutation relations can be used to find the spherical equivalents of cartesian tensors. If $\mathbf{A}$ is a vector, then $[J_z, A_z] = 0$ and so $A_0 = A_z$. One then finds $A_{\pm 1} = \frac{1}{\sqrt{2}}[J_\pm, A_0] = \mp \frac{1}{\sqrt{2}}(A_x \pm iA_y)$.

Writing the dipole operator as a scalar product of irreducible tensors allows to nicely separate the geometric part of the problem (dependence on magnetic quantum numbers) from the excitation (E-field: π -polarized, σ_+ -polarized etc.).

For more information on the general topic of spherical tensors, irreducible representations of the rotation group, Wigner D-matrices, etc., please see:

[1] D. Brink, G. Satchler, Angular Momentum (Clarendon Press 1968) Very clear, concise, thorough presentation.

[2] Julian Schwinger, On Angular Momentum, 1952 Builds up general angular momentum from spin 1/2 particles, in an elegant, powerful treatment. For that latter technique, also see

[3] F. Bloch, I.I. Rabi, Atoms in Variable Magnetic Fields, Rev. Mod. Phys. 17, 237 (1945). They cite a paper by E. Majorana, Nuovo Cimento 9, 43 (1932) which is the first to present the so-called "Majorana star" representation of an angular momentum state.